

**Problem Set 1**  
**Due: Mar. 1, 2021**

1. Section 1.2 Problem 7.

A four-sided die is rolled repeatedly, until the first time (if ever) that an even number is obtained. What is the sample space for this experiment?

**Solution:** The outcome of this experiment can be any finite sequence of the form  $(a_1, a_2, \dots, a_n)$ , where  $n$  is an arbitrary positive integer,  $a_1, a_2, \dots, a_{n-1}$  belong to  $\{1, 3\}$ , and  $a_n$  belongs to  $\{2, 4\}$ . In addition, there are possible outcomes in which an even number is never obtained. Such outcomes are infinite sequences  $(a_1, a_2, \dots)$ , with each element in the sequence belonging to  $\{1, 3\}$ . The sample space consists of all possible outcomes of the above two types.

2. Section 1.2 Problem 8.

You enter a special kind of chess tournament, in which you play one game with each of three opponents, but you get to choose the order in which you play your opponents, knowing the probability of a win against each. You win the tournament if you win two games in a row, and you want to maximize the probability of winning. Show that it is optimal to play the weakest opponent second, and that the order of playing the other two opponents does not matter.

**Solution:** Let  $p_i$  be the probability of winning against the opponent played in the  $i$ th turn. Then, you will win the tournament if you win against the 2nd player (probability  $p_2$ ) and also you win against at least one of the two other players [probability  $p_1 + (1 - p_1)p_3 = p_1 + p_3 - p_1p_3$ ]. Thus, the probability of winning the tournament is

$$p_2(p_1 + p_3 - p_1p_3).$$

The order  $(1; 2; 3)$  is optimal if and only if the above probability is no less than the probabilities corresponding to the two alternative orders, i.e.,

$$p_2(p_1 + p_3 - p_1p_3) \geq p_1(p_2 + p_3 - p_2p_3),$$

$$p_2(p_1 + p_3 - p_1p_3) \geq p_3(p_2 + p_1 - p_2p_1).$$

It can be seen that the first inequality above is equivalent to  $p_2 \geq p_1$ , while the second inequality above is equivalent to  $p_2 \geq p_3$ .

3. Let  $\{A_i : i \in I\}$  be a collection of sets. Prove the "De Morgan's Law":

$$\left(\bigcup_i A_i\right)^c = \bigcap_i A_i^c, \quad \left(\bigcap_i A_i\right)^c = \bigcup_i A_i^c.$$

**Solution:**

- (a) Let  $a \in (\bigcup A_i)^c$ . Then  $a \notin \bigcup A_i$ , so that  $a \in A_i^c$  for all  $i$ . Hence  $(\bigcup A_i)^c \subseteq \bigcap A_i^c$ . Conversely, if  $a \in \bigcap A_i^c$ , then  $a \notin A_i$  for every  $i$ . Hence  $a \notin \bigcup A_i$ , and so  $\bigcap A_i^c \subseteq (\bigcup A_i)^c$ . The first De Morgan's law follows.
- (b) Applying part (a) to the family  $\{A_i^c : i \in I\}$ , we obtain that  $(\bigcup_i A_i^c)^c = \bigcap_i (A_i^c)^c = \bigcap_i A_i$ . Taking the complement of each side yields the second law.
4. Six cups and saucers come in pairs: there are two cups and saucers which are red, two white, and two with stars on. If the cups are placed randomly onto the saucers (one each), find the probability that no cup is upon a saucer of the same pattern.

**Solution:** Lay out the saucers in order, say as RRWWSS. The cups may be arranged in  $6!$  ways, but since each pair of a given color may be switched without changing the appearance, there are  $6! \div (2!)^3 = 90$  distinct arrangements. By assumption these are equally likely. In how many such arrangements is no cup on a saucer of the same color? The only acceptable arrangements in which cups of the same color are paired off are WWSSRR and SSRRWW; by inspection, there are a further eight arrangements in which the first pair of cups is either WS or SW, the second pair is either SR or RS, and the third pair either RW or WR. Hence the required probability is  $10/90 = \frac{1}{9}$ .